EXPERIMENT-12

End –End Communication at Transport Layer

a) Implement echo client server using TCP/UDP sockets.
b) Implement a chat program using socket programming.

tcp_server.py import socket server =

socket.socket(socket.AF_INET, socket.SOCK_STREAM)

server.bind(("127.0.0.1", 5000)) server.listen(1) print("TCP

Echo Server started...") conn, addr = server.accept()

print("Connected:", addr)

while True:

data = conn.recv(1024) if not

data: break

print("Client:", data.decode())

conn.send(data)

conn.close() server.close()

tcp_client.py import

socket

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

client.connect(("127.0.0.1", 5000))

print("Connected to TCP Echo Server")

while True:

msg = input("Enter message: ")

if msg.lower() == "exit":

break

client.send(msg.encode())

data = client.recv(1024)

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```
    print("Server:", data.decode())
client.close()
```

udp_server.py import socket

```
server = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
server.bind(("127.0.0.1", 5001))
print("UDP Echo Server started...")
while True:
    data, addr = server.recvfrom(1024)
print("Client:", data.decode())
    if data.decode().lower() == "exit":
        break
    server.sendto(data, addr)
server.close()
```

udp_client.py import

socket

```
client = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
print("Connected to UDP Echo Server")
while True:
    msg = input("Enter message: ")
    client.sendto(msg.encode(), ("127.0.0.1", 5001))
    if msg.lower() == "exit":
        break
    data, addr = client.recvfrom(1024)
print("Server:", data.decode())
client.close()
```

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chat_server.py import socket server = socket.socket(socket.AF_INET, socket.SOCK_STREAM) server.bind(("127.0.0.1", 5002)) server.listen(1)

print("Chat Server started...") conn, addr = server.accept()

print("Connected:", addr) while True:

msg = conn.recv(1024).decode()

if msg.lower() == "exit":

break

print("Client:", msg)

reply = input("You: ")

conn.send(reply.encode())

if reply.lower() == "exit":

break

conn.close() server.close()

chat_client.py import

socket

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

client.connect(("127.0.0.1", 5002))

print("Connected to Chat Server")

while True:

msg = input("You: ")

client.send(msg.encode())

if msg.lower() == "exit":

break

reply = client.recv(1024).decode()

print("Server:", reply)

if reply.lower() == "exit":

break client.close()